Numerical Analysis Exercise 4

4.Use the Gaussian elimination with backward substitution to solve the following liner systems.

$$
(1)\begin{cases} 4x_1 - x_2 + 3x_3 = 8 \\ 2x_1 + 5x_2 + 2x_3 = 3. \\ x_1 + 2x_2 + 4x_3 = 11 \end{cases}
$$

Solution

```
clc
clear
%Gaussian Elimination Method%
A=input("Please input the a:")
b=input("Please input the b:")
b=b';
Gauss(A,b)
```

```
function X=Gauss(A,b)
ab=[A,b];%a为系数矩阵，b为右端项
[R,C]=size(ab);
for i=1:R-1 %控制消去首行元素的次数 二阶方程组只需要消去一次,三阶消去两次 i=2时，系数矩阵第三行消去一个首行元素,此时系数矩阵第三行只有一个元素
    for j=i+1:R         %假设R=3，i=1时，j=2.此时内循环两次，系数矩阵的第二三行分别消去一个首行非零元素
        ab(j,i:C)=ab(j,i:C)-ab(i,i:C)*ab(j,i)/ab(i,i);%每一行消去首行元素
    end
end
x=zeros(R,1);%解空间
x(R,1)=ab(R,C)/ab(R,R);

fprintf("解为: ")
for k=R-1:-1:1%假设R=3，C=4; k从2开始循环
    x(k)=(ab(k,C)-ab(k,k+1:R)*x(k+1:R))/ab(k,k);%反向替换
end
x
end
```

```
A = 3×3
     4    -1     3
     2     5     2
     1     2     4
```

```
b = 1×3
      8      3     11

R = 3
C = 4
解为:
x = 3×1
   -0.4328
   -0.4627
    3.0896
```

$$(2)\begin{cases} 4x_1+x_2+3x_3=9\\ 2x_1+4x_2-x_3=-5.\\ x_1+2x_2-3x_3=-9 \end{cases}$$

```
A = 3×3
      4      1      2
      2      4     -1
      1      1     -3

b = 1×3
      9     -5     -9

解为:
x = 3×1
      1
     -1
      3
```

5.Find the parabola $y=Ax^2+Bx+C$ that passes throgh the points (1,4),(2,7) and (3,14).

Solution Since the curve passer throgh these points,so we plug the points into the equation,

$$\begin{cases} A+B+C=4\\ 4A+2B+C=7.\\ 9A+3B+C=14 \end{cases}$$

By using the model in the fourth question,we have,

```
A = 3×3
      1      1      1
      4      2      1
      9      3      1

```

```
b = 1×3
     4     7    14

解为:
x = 3×1
     2
    -3
     5

```

So ,we have the A = 2, B = -3 , C =5 . thus, we use the matlab figuring